→ where AI is used in software Domain

1) AI - Based Bug Detection

- Agent : Bug Detection AI
- Perception : Source code
- cognition : ML Pattern analysis
- Action / output : Detects Bugs

2) Intelligent code Auto completion

- Agent : code completion AI
- Perception : current code context
- cognition : Deep learning model
- Action / output : Predicts and completes code

3) AI Code Review system

- Agent : code review AI
- Perception : Pull request code
- cognition : static analysis + ML
- Action / output : Gives review comments.

4) AI software testing

- Agent : Testing AI
- Perception : APP behaviour
- cognition : Reinforcement learning
- Action / output : Auto Generates test cases

5) AI Requirement analysis

Agent: Requirement AI
Perception: Client documents
Cognition: NLP
Action/output: Extracts requirements

6) AI Project Time Estimation

Agent: Project AI
Perception: Past project Data
Cognition: predictive ML
Action/output: Estimates timelines

7) AI DevOps monitoring

Agent: Monitoring AI
Perception: Logs / system metrics
Cognition: Anomaly detection
Action/output: sends alerts

8) AI Documentation Generator

Agent: Documentation AI
perception: source code
cognition: NLP
Action/output: Generates Documentation.

9) AI API Recommendation

Agent: Recommendation AI
Perception: Project requirements.

Cognition: Recommender System
Action/output: suggests ui

10) AI security vulnerability Detection

Agent: security AI
Perception: code & network Data
Cognition: ML security model
Action/output: Detects vulnerability.